Quantifying similarity between copy number profiles with CNVMetrics package

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ABSTRACT

Genome-wide DNA copy number profiles are an informative type of molecular data that are used in numerous areas of genomic analysis and can be derived from a variety of platforms, including microarrays and next-generation DNA sequencing. Copy-number variants (CNVs) have been shown to be associated with a wide spectrum of pathological conditions and complex traits. They are also ubiquitously present and predictive of patient outcomes in cancer.

While visual representation is a practical way to qualitatively compare copy number profiles, metrics are useful statistical tools for quantitatively measuring similarity and dissimilarity between profiles. Similarity metrics can be employed to compare CNV profiles of genetically unrelated samples as well as those with a common genetic background. As an example, a comparison between primary and metastatic tumor CNV profiles may reveal genomic determinants of metastasis. Similarly, for patient-derived xenograft or organoid models, *CNVMetrics* can quantify the recapitulation of the CNV patterns of the tumor tissue of origin.

The *CNVMetrics* package offers multiple quantitative metrics of similarity between copy number profiles. Among these are metrics based on CNV categorical status calls or on the copy number ratios. The input is a plain text file containing the genomic position of the copy number variations, as well as the status and/or the log2 ratio values. *CNVMetrics* is using data structures supplied by the Bioconductor GenomicRanges package to store and manipulate the CNVs.

Significance of the observed metrics is assessed in comparison to the null distribution, computed using simulated profiles. Functions implementing the simulation methods are included in the package.

Moreover, a visualization tool is also provided to explore resulting metrics in the form of profile-to-profile heatmaps.

As an illustration, we describe an application of *CNVMetrics* to CNV profiles of pancreatic cancer specimens and organoids derived therefrom.

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Introduction

The copy number variant (CNV) profiles from derived preclinical disease models (xenografts, organoids, etc.) must recapitulate the genomic features of the primary tissues. While visual representation is a practical way to qualitatively compare CNV profiles, metrics provide quantitative measures.

The *CNVMetrics* package implements multiple quantitative metrics of similarity between CNV profiles. Among these are metrics of similarity based on CNV categorical status calls or on the copy number ratios. A simulation method that enables p-value calculation for those metrics is also included. In addition, a visualization tool is provided to explore resulting metrics in the form of profile-to-profile heatmaps.

Method

Input

This column that identifies the samples is **mandatory**. However, it can be located at any position and have a user-specific name.

The columns that provide the **genomic location** of the CNV calls are **mandatory**.

This column is required by metrics using the level of amplification/deletion and for the simulation step.

ID	chr	start	end	state	log2ratio
mouse1	chr1	3527631	95471971	NEUTRAL	0.00635
mouse1	chr2	3549741	182113224	AMPLIFICATION	0.04883
mouse1	chr3	3536052	127918384	DELETION	-0.71932
mouse2	chr1	3527655	195471955	NEUTRAL	0.01123
mouse2	chr2	1431113	91533212	AMPLIFICATION	0.43321
mouse3	chr1	2941755	112113212	NEUTRAL	0.01212
mouse3	chr2	2231131	100132241	AMPLIFICATION	0.89123
...					

This column is required by metrics using CNV status calls. The recognized values are:
- **AMPLIFICATION**
- **DELETION**

Other values are ignored.

For the simulation step, all segments should be present, including **NEUTRAL** segments.

Figure 1. Example of a *CNVMetrics* input file. The **BED format** file contains the position of the copy number events, the state and/or the level of amplification. The names of the columns should be respected. All samples are identified by an unique ID.

Workflow

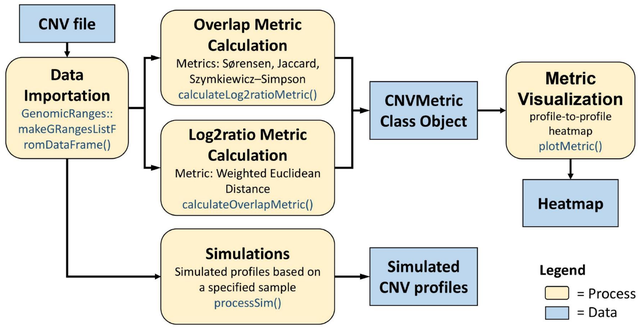

Figure 2. From a standard input based on *GenomicRanges*[1], we implement multiple quantifications of the similarity between copy number variation profiles and an overall visualization tool. A simulation method is also available.

Metrics

Status calls (amplification/deletion)		
Sørensen	**Szymkiewicz–Simpson**	**Jaccard**
$\frac{2 \times \lvert X \cap Y \rvert}{\lvert X \rvert + \lvert Y \rvert}$	$\frac{\lvert X \cap Y \rvert}{min(\lvert X \rvert, \lvert Y \rvert)}$	$\frac{\lvert X \cap Y \rvert}{\lvert X \cup Y \rvert}$
Where X and Y represent the regions of each sample in base paired.	Where X and Y represent the regions of each sample in base paired. If set X is a subset of Y or vice versa, the overlap coefficient value is 1.	Where X and Y represent the regions of each sample in base paired.

Table 1. Metrics using the status calls[2,3,4].

Level of amplification or deletion (log2 ratio)
Weighted Euclidean Distance-Based
$\frac{1}{1 + \sqrt{\sum_{i=1} log_2(w_i)(A_i - B_i)^2}}$
Where A_i and B_i represent the log2 ratio values of samples A and B for the region *i* while w_i is the length of region *i* in base paired.

Table 2. Metric using the level of amplification/deletion.

Simulations

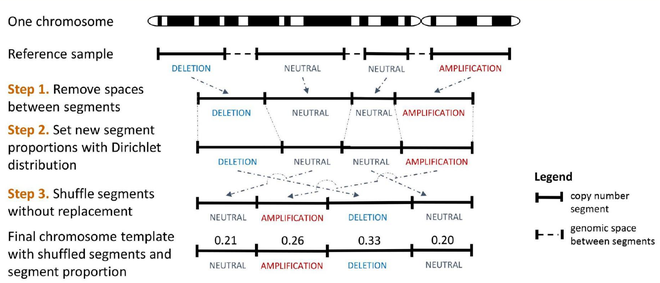

Figure 3. Workflow showing the creation of a chromosome template that is used to simulate chromosomes.

Simulations

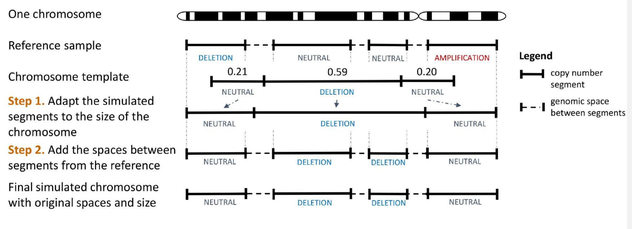

Figure 4. Workflow showing the creation of a simulated chromosome through the use of a template.

Results – Real data

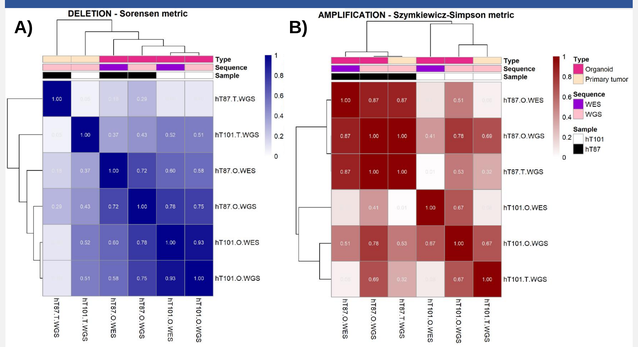

Figure 4. Heatmaps based on overlap metrics done on pancreatic tumor organoids with matching primary[5]: Sørensen metrics for deleted regions (A); Szymkiewicz-Simpson metrics for amplified regions (B). Alignment was done with BWA[6] v0.7.17; CNV were identified with PureCN[7] v1.18.1 (WES) and FACET[8] v0.6.1 (WGS).

Conclusion

- The metrics enable a quantitative comparison of CNV profiles.
- The *CNVMetrics* package is part of Bioconductor https://bioconductor.org/packages/CNVMetrics
- The simulation method will be part of the next release.

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